Unit: Unit 3: Functions **Name:** ____________
Topic: Definition of Relation and Function, Domain, Range, and Notations **Date:** __________

Advanced (Habanero)

Q1. What is the relation between x and y in the equation $y = \mathrm{x}^2$?
(A) Linear
(B) Quadratic
(C) Cubic
(D) Exponential
(E) Logarithmic

Q2. Which of the following is NOT a function?
(A) $f(x) = x$
(B) $f(x) = \mathrm{x}^2$
(C) $f(x) = |x|$
(D) $f(x) = 1\frac{}{x}$
(E) $f(x) =$x, y, x=y

Q3. What is the domain of the function $f(x) = 1\frac{}{x}$?
(A) $(-\infty, -1]$
(B) $(-\infty, 0)$
(C) $(-\infty, 0) \cup (0, \infty)$
(D) $[0, \infty)$
(E) $(-1, 1)$

Q4. Given the function $g(x) = 2x - 1$, what is $g(3)$?
(A) 5
(B) 6
(C) 7
(D) 8
(E) 9

Q5. What is the range of the function $h(x) = x^2 + 2$?
(A) $[0, \infty)$
(B) $[2, \infty)$
(C) $(-\infty, 2]$
(D) $(-2, 2)$
(E) $(-\infty, 0)$

Q6. Which of the following relations is a function?
(A) $f(x) = x, y, x = y$
(B) $x = 2y$
(C) $x^2 + y^2 = 4$
(D) $y = 1\frac{}{x}$
(E) $x + y = 3$

Q7. Given the function $f(x) = x^2 - 4x + 3$, what is the value of $f(-2)$?
(A) -9
(B) -5
(C) 3
(D) 7
(E) 11

Q8. What is the equation of the vertical line that passes through the point $(2, 3)$?
(A) $x = 2$
(B) $y = 3$
(C) $y = 2x - 1$
(D) $x + y = 5$
(E) $x - y = -1$

Open Ended Questions (FRQ)

Q9. Determine the domain and range of the function $f(x) = \sqrt{x+1}$ and explain why.

Q10. Given the relation $x^2 + y^2 = 9$, determine whether this relation is a function or not and explain why. Also, graph the relation.

Chef's Tips

- Tip 1: Check for any calculations or graphing errors
- Tip 2: Make sure students provide adequate explanations for FRQs
- Tip 3: Ensure students use proper function notation